

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI / ML engineer shipping production healthcare AI end to end: HIPAA-conscious RAG (~35% retrieval precision, >90% grounded alignment), predictive ML pipelines (15-20% recall on high-risk patient categories at >90% precision), agentic LLM workflows (>30% hallucination reduction), and FastAPI + Docker packaging (~30% post-deploy defect reduction). Production computer vision via YOLOv8 (~30% inference latency reduction). Master's in Computer Science, Kent State. Targeting R37 Lab / R1 RCM's ML Engineer seat (NYC | Remote-US, AI-native Healthcare Revenue Operating System at 95 of top 100 US health systems) - production healthcare RAG with retrieval/ranking/categorization/generative AI over EHR (1:1 with R37's "reason over full medical records, payer logic, and financial workflows" charter), Agentic Healthcare Claims with five schema-validated agents and RAG-grounded CPT/ICD policy retrieval (matches automate medical coding + billing + follow-up), HIPAA-conscious data governance + evaluation audit trails (matches scientific rigor + audit-ready reasoning), and Manga Lens 4-provider AI vision shipped to Chrome Web Store (proves 0 → 1 + production hardening). L2-fit per ladder definition (independent end-to-end ownership). NYC relocation open, F-1 OPT US-base, US-domestic travel for hub interview OK.

CORE COMPETENCIES

ML Engineer (Applied AI Systems) • Production Healthcare RAG / Embeddings / Vector Search • Agentic Pipelines / Schema Contracts / Audit Trails • Predictive ML (scikit-learn, XGBoost) / Threshold Calibration • Evaluation Pipelines / Benchmarking / Experiment Tracking • Multi-Provider LLM API (OpenAI, Anthropic, Google, Ollama) • FastAPI / Flask / Docker / REST • CI/CD (Jenkins, GitHub Actions) / Grafana • HIPAA-Conscious Data Governance • 0 → 1 Founder / Public Ship Discipline • NYC Relocation Open (F-1 OPT US-base)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Applied AI, Healthcare

USA - Healthcare Technology, Applied AI

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved retrieval precision ~**35%**; retrieval-grounded response alignment >**90%**.
- Designed **retrieval / ranking / categorization** pipelines over unstructured EHR + appointment + ticket data; instrumented evaluation datasets and validation gates tied to clinical outcomes (recall, precision, grounded alignment).
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding; agent stability ~**25%**; hallucinations >**30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement scoring, support prioritization; recall +**15-20%** on high-risk categories at >**90%** precision via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~**30%**.
- Authored **HIPAA-conscious data governance**, lineage, evaluation audit trails, and stakeholder-facing system-limitation docs; ran weekly stakeholder calls with clinical SMEs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP

- Optimized SQL and **T-SQL stored procedures** on the Synthesis Order-to-Cash platform; query execution time **~20%**, data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; deployment errors **>30%**, release cycle **~35-40%**.
- Designed **SQL Server DMVs and Grafana** performance dashboards for long-running queries, deadlocks, CPU/IO contention; incident recurrence **~25%**, RCA speed **~30%**.
- Implemented **RBAC and audit logging for financial modules** in a compliance-sensitive oil & gas environment.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization + timestamp-aligned clip extraction cut manual review **60-70%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Five-stage multi-agent pipeline (Intake Normalization, Validation, Consistency Analysis, Duplicate Detection, Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. **RAG-grounded CPT/ICD validation** against vector-indexed policy documents. Approximate-nearest-neighbor similarity search for duplicate claim detection. Audit-ready explainable risk scoring with reasoning traces.

Python, LangChain, FastAPI, vector search, GPT-4, schema contracts, ANN, audit trails

Manga Lens - Multi-Provider AI Vision Chrome Extension (shipped, Chrome Web Store)

Real-time manga / webtoon translation extension shipped to the Chrome Web Store. **Multi-provider abstraction layer** integrating four AI vision providers (Claude Sonnet, GPT-4o-mini, GPT-4.1-Nano, Ollama/minicpm-v) with per-provider payload handling (WebP for cloud, JPEG for local), cost-aware routing, circuit-breaker fallback. Manifest V3 service workers; multi-section capture with coordinate remapping; 7-day cache; per-domain selector configs for 29 sites; narrowed host permissions and privacy policy.

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / Ollama, multi-provider integration, cost-aware routing

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. **Stable Diffusion fine-tuning + LoRA** for identity, **ControlNet + OpenPose / MediaPipe** for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks.

PyTorch, Hugging Face Diffusers, Stable Diffusion, LoRA, ControlNet, OpenPose, FastAPI, Docker

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** detection-and-classification model; reduced inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives **~25%**; adaptive frame skipping; NMS tuning for stable real-time operation via OpenCV.

YOLOv8, OpenCV, PyTorch, real-time inference

E-Farming Digital Marketplace (Founder & Full-Stack Developer)

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded **80-120 active users** in phase one; cart, reviews, and community blogging features.

PHP, MySQL, Bootstrap, full-stack, founder-led

EDUCATION

Master of Science, Computer Science - Kent State University	Aug 2023 - May 2025
Focus: Applied Machine Learning, NLP, Database Systems.	
Bachelor of Technology, Computer Science - KL University	Jun 2019 - May 2023
Founder / Lead of E-Farming platform (university project).	

CERTIFICATIONS

Deep Learning Specialization - <i>DeepLearning.AI</i> (Coursera)	2024
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SKILLS

LLM / GenAI: RAG (recursive semantic chunking, embeddings, vector search), agentic workflows, multi-step agents, schema-validated JSON contracts, structured outputs, evaluation pipelines, guardrails, grounding, LangChain, multi-provider LLM API (OpenAI, Anthropic, Google, Ollama)

ML Frameworks: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LoRA fine-tuning

Languages: Python (primary), TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++

Backend / Inference: FastAPI, Flask, REST, Docker, structured logging, load simulation, microservices

Data & Evaluation: Pandas, NumPy, feature engineering, threshold calibration, class weighting, stratified sampling, evaluation pipelines, audit trails, benchmarking

DevOps: Docker, Jenkins, GitHub Actions, Grafana, CI/CD, cloud-ready deployment

Computer Vision: YOLOv8, OpenCV, ControlNet, OpenPose / MediaPipe, Stable Diffusion + LoRA

Data Governance: HIPAA-conscious de-identification, data lineage, evaluation audit trails, RBAC, audit logging

Domains: Healthcare AI / RCM-adjacent, Production AI Agents, Founder / 0-to-1, Compliance-Sensitive Integration